

Counting and combinations

How many ways can it happen?

THE IDEA

Probability needs counting: how many outcomes are there in total, and how many are favourable? Three tools cover almost everything.

Tool	Formula	Use when
Multiplication rule	$a \times b$	Choices happen in stages
Factorial	$n! = n \times (n-1) \times \dots \times 1$	Arranging all n items
Permutation	$n! / (n-r)!$	Picking r of n and order matters
Combination	$n! / [r! (n-r)!]$	Picking r of n and order does not matter

The only question to ask: does the order matter? First and second place in a race — yes, permutation. Two people on a team — no, combination. Note $0! = 1$.

WORKED EXAMPLES

TOPIC 4 — COUNTING & COMBINATIONS

how many ways can it happen ?

THE MULTIPLICATION RULE

choices in stages → MULTIPLY them

$$3 \text{ shirts} \times 4 \text{ trousers} = 12 \text{ outfits}$$

FACTORIAL $n!$

$n! = n \times (n-1) \times \dots \times 1$ arrangements of n items

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

note : $0! = 1$

PERMUTATION — order MATTERS

$$P(n, r) = n! / (n - r)!$$

example : 1st and 2nd place from 5 runners

$$5! / 3! = 120 / 6 = 20$$

COMBINATION — order does NOT matter

$$C(n, r) = n! / [r! \cdot (n - r)!]$$

example : choose 2 people from 5 for a team

$$5! / (2! \cdot 3!) = 120 / (2 \times 6) = 10$$

half of the permutation answer — because AB and BA are the same team but different race results.

THE ONE QUESTION TO ASK : does the ORDER matter ?

yes → permutation. no → combination.

Notice the relationship: the combination answer is always the permutation answer divided by $r!$, because each unordered group has $r!$ different orderings that all count as the same choice.

Why it matters for ML: counting underlies the binomial distribution, cross-validation splits, sampling strategies, and how many possible hyperparameter combinations a grid search must try.

KEY POINTS FROM THIS TOPIC

Multiplication	choices in stages → multiply
Factorial	$n! = n \times (n-1) \times \dots \times 1$, $0! = 1$
Permutation (order matters)	$n! / (n-r)!$
Combination (order does not)	$n! / [r! (n-r)!]$

TEST — Topic 4

1. Compute $4!$ → **24**
2. From 6 people, how many ways to choose a team of 2 (order does not matter)? → **$6!/(2! \cdot 4!) = 720/(2 \cdot 24) = 15$**
3. From 6 runners, how many ways to award 1st and 2nd place (order matters)? → **$6!/4! = 720/24 = 30$**